The Golden Ladder

Golden Cuts from Plato to the Fourth Dimension

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Chapter 1, *One Triangle, Three Numbers,* opens the book; chapter 2, *Silence Is Golden,* follows. Later chapters are added in new versions.

Chapter 1 · One Triangle, Three Numbers

The five Platonic solids come from three kaleidoscopes, and the three kaleidoscopes from one triangle with a single number in its corner; that number, 1, √2 or φ, decides where every golden cut of this book can and cannot live

DICTO proposed that the Platonic solids reduce further than five, that a dual pair is in essence one shape, that the cube and the octahedron pair space between them, and that the stella octangula and the Dogstar tie the solids together; he also proposed that, on this basis, most of the book's calculations are already done and what remains is arrangement and interpretation. Drafting with Claude (Anthropic).

> *Hold two mirrors at an angle and a single bead becomes a ring of beads. Hold three mirrors together like the corner of a box and a bead becomes a whole solid. Every Platonic solid is made this way, by one small triangle of mirrors, and only three such triangles exist. Each carries one number: 1, √2 or the golden ratio φ. This book climbs from that triangle to the fourth dimension, and that number decides at every step which golden shapes can exist.*

1. Five solids, three families

The five Platonic solids, in DICTO order, are the dodecahedron, the icosahedron, the cube, the tetrahedron and the octahedron. They are not five unrelated things. Put a point at the centre of every face of a cube and join neighbouring points: you get an octahedron. Do the same to the octahedron and you are back at the cube. The dodecahedron and the icosahedron are partners in the same way, and the tetrahedron is its own partner. These partners are the **duals**, and a dual pair is, in essence, one shape seen from two sides: the two have exactly the same mirrors (nine planes for the cube and octahedron, fifteen for the dodecahedron and icosahedron, six for the tetrahedron), and the corners of one point exactly where the faces of the other point. Where a cube and its octahedron overlap you get the cuboctahedron; the hull of the two is Kepler's rhombic dodecahedron.

So the five solids are really three families: the **dodecahedron–icosahedron** pair, the **cube–octahedron** pair, and the **tetrahedron**, which pairs with itself. When a tetrahedron and its own dual are put together they make the **stella octangula**: its hull is a cube and its core an octahedron. So the tetrahedron's family already holds the cube and the octahedron inside it.

2. One triangle of mirrors

Each family has a set of mirror planes through its centre. They cut the surface of a sphere into small identical triangles, and every one of these triangles has the same three angles: a right angle, 60°, and 180°/p, with p = 3 for the tetrahedron, p = 4 for the cube and octahedron, and p = 5 for the dodecahedron and icosahedron (Figure 1). Each triangle is a kaleidoscope: a single point inside it, reflected in its three walls again and again, makes the corners of a whole solid. This is Wythoff's construction (1918), set out by Coxeter in *Regular Polytopes*.

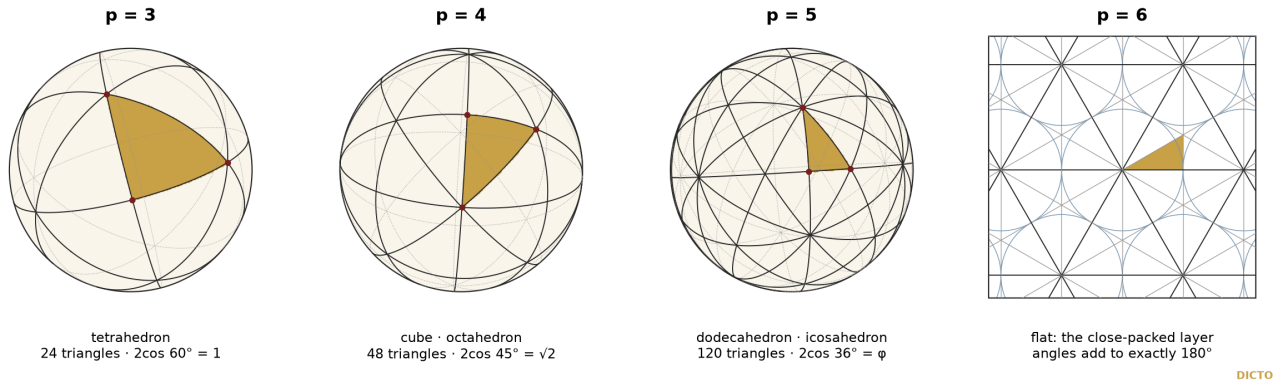


Figure 1. The mirror planes of the three families cut the sphere into 24, 48 and 120 copies of one triangle (gold), with angles 90°, 60° and 180°/p, p = 3, 4, 5; red dots mark its corners. At p = 6 the angles add up to exactly 180° and the triangle is flat: it tiles the plane under the close-packed layer of balls (blue circles). Below each picture, the number 2 cos(180°/p) of that kaleidoscope.

Where the point sits decides the solid. In the cube's triangle (p = 4) the three corners lie on the three kinds of axis: a point at the four-fold corner gives the **octahedron**, a point at the three-fold corner gives the **cube**, and a point at the right-angled corner gives the **cuboctahedron**. Slide the point along the side from the octahedron's corner to the cuboctahedron's and you pass through every octahedron with its corners cut, the truncated octahedron among them; slide on from the cuboctahedron's corner to the cube's and you pass through every cube with its corners cut, the truncated cube among them. That walk, octahedron → cuboctahedron → cube along two sides of one triangle, is exactly the sweep of paper #90 through the six-dimensional cube × octahedron, and the weight walk by which the bond papers (#48, #53) shared out space between two kinds of cell. In the p = 5 triangle the same walk runs icosahedron → icosidodecahedron → dodecahedron, and in the p = 3 triangle tetrahedron → octahedron → the other tetrahedron. Points on the third side and inside the triangle give most of the other Archimedean solids; the two snub solids come from a variant of the same construction.

So the five solids reduce to three triangles, and the three triangles to one shape with one free number p. Every **golden cut** of this book (#88) is a point on a side of one of these triangles: the corner of the solid is cut back until each face becomes a truncated polygon whose sides are in the golden ratio.

3. Why only three, and the three numbers

A triangle drawn on a sphere has angles adding up to more than 180°, and the excess is its area. With angles 90°, 60° and 180°/p, the excess is 180° × (1/2 + 1/3 + 1/p − 1), which is positive only for p = 3, 4, 5. That is the whole reason there are exactly five Platonic solids. The number of triangles on the sphere is 720°/excess: 24, 48 and 120.

Each triangle carries one number, 2 cos(180°/p), the length that its corner angle gives to everything built in it:

p	family	2 cos(180°/p)	its world of numbers
3	tetrahedron	1	whole numbers and fractions
4	cube · octahedron	√2	the silver ratio 1 + √2
5	dodecahedron · icosahedron	φ = (1 + √5)/2	the golden ratio
6	flat: the close-packed layer	√3	the plane, no longer a sphere

These are the same numbers as the $c_n = 2\cos(180°/n)$ in the formula for the golden cuts of #88, where n is the number of sides of a face: 1 for the triangle faces, √2 for the cube's squares, φ for the dodecahedron's pentagons. That is why the golden cuts of the tetrahedron, octahedron, icosahedron and dodecahedron all live with √5 alone, while the two Golden Cut Cubes need √2 and √5 together, and why no golden structure in any dimension holds them: the field obstruction of #90. The golden ratio belongs to p = 5; the cube's own number is √2. The tetrahedron, with the plain number 1, belongs to both.

4. The cube in the middle, and the two stars

The tetrahedron's turning symmetry sits inside both larger families, and the cube is where they meet. Two tetrahedra fit in a cube, corner to corner, making the stella octangula. A cube fits in a dodecahedron, corner to corner, five ways, and Euclid built the dodecahedron exactly so: a cube with a roof on each of its six faces (Elements XIII.17). Take the roofs away from the inside instead and you are left with the **Dogstar**, the cube minus six roofs reaching into it: Hart's stellation 8 of the dodecahedron, which fills space together with regular dodecahedra (#42). So the two stars are the cube's two ties: the stella octangula ties it to the tetrahedron and so to the octahedron, and the Dogstar ties it to the dodecahedron.

The cube and the octahedron also pair space between them. Cut a cube by an octahedron at any depth; the cut solid at the partner depth fills the gaps, so the two always fill space together, the two kinds of cell of the caesium chloride crystal (#90, Proposition 6). At one end the octahedron's share vanishes and the cube fills space alone; in the middle the cuboctahedron pairs with the octahedron; beyond it every corner-cut cube pairs with a whole octahedron. Together with the tetrahedron, which fills space with the octahedron in the octet (#44, #92), the cube's family is the one that tiles. The golden family does not: the dodecahedron and the icosahedron need partners from outside the five.

5. The flat triangle: where the stacking papers live

At p = 6 the angles 90°, 60° and 30° add up to exactly 180°, and the triangle lies flat. It no longer closes up into a sphere; it tiles the plane, and the mirrors it makes are those of a flat layer of balls packed as tightly as possible. Stack such layers and you have every close-packed crystal, straight (cubic) or twisted (hexagonal) at each layer: the setting of the Hexagonality Law and the stacking papers, and of #92, where the tilings by the octet's pyramids turn out to be exactly these stackings. So the series stands on the two sides of one boundary: the spherical triangles, p = 3, 4, 5, which make the solids, and the flat triangle, p = 6, which makes the layers.

6. The ladder

DICTO proposed that on this basis most of the book's calculations are already done, and that what remains is arrangement and interpretation. The triangle says which shapes can exist; the papers of the series measured where along its sides the special points fall: the golden cuts (#88), their homes in four dimensions (#89), why the cube's pair has none there (#90), and the four-dimensional solids whose every cell is golden (#91). The chapters that follow climb that ladder one rung at a time:

rung	chapter	papers
2D	the truncated triangle, and the golden bond	#20, #48
3D	Every Platonic Solid Has Exactly Two Golden Cuts	#88
4D	the 120-cell atlas	#65
4D	All Five Golden Pairs Are Slices of Four-Dimensional Solids	#89
6D	Golden on the Octahedron, Silver on the Cube	#90
4D	Four-Dimensional Solids Whose Every Cell Is a Golden Cut	#91
8D	the Golden Ladder: E8 and its golden twist	closing chapter

At every rung the question is the one this chapter answers for three dimensions: which of the three numbers, 1, √2 or φ, does the shape need, and does the dimension we are in supply it?

Sources

Euclid, *Elements*, tr. T. L. Heath, Book XIII, Proposition 17 (the dodecahedron as a cube with six roofs). W. A. Wythoff, "A relation between the polytopes of the C600-family", Proc. KNAW 20 (1918). H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover (1973), chapters 4–5 (kaleidoscopes, the triangles (2, 3, p)). G. W. Hart, Tetrahedral stellations of the dodecahedron (1996), stellation 8. DICTO, The DICTO Sunstar, #42, DICTO DOI: 10.5281/zenodo.23497702. DICTO, Aristotle's Pyramid, #44, DICTO DOI: 10.5281/zenodo.23529497. DICTO, The Golden Split, #48, DICTO DOI: 10.5281/zenodo.23596362. DICTO, The Global Web of Bonds, #53, DICTO DOI: 10.5281/zenodo.23631140. DICTO, Every Platonic Solid Has Exactly Two Golden Cuts, #88, DICTO DOI: 10.5281/zenodo.23896439. DICTO, All Five Golden Pairs Are Slices of Four-Dimensional Solids, #89, DICTO DOI: 10.5281/zenodo.23917931. DICTO, Golden on the Octahedron, Silver on the Cube, #90, DICTO DOI: 10.5281/zenodo.23931136. DICTO, Four-Dimensional Solids Whose Every Cell Is a Golden Cut, #91, DICTO DOI: 10.5281/zenodo.23931389. DICTO, Aristotle's Pyramid in Layers, #92, DICTO DOI: 10.5281/zenodo.23931608.

Chapter 2 · Silence Is Golden

The sweep from the octahedron to the cube is the family of weight shapes of the particle symmetries SU(4) and SO(7), and in four dimensions of SO(9); every golden solid of the sweeps is where a Fibonacci ladder of multiplets converges, and the first half of the sweep is one musical octave

DICTO proposed widening the search to the particle zoo, harmonics and tuning (The DICTO Concordance, Part 5), and making these results a chapter of The Golden Ladder. Computation and drafting with Claude (Anthropic).

> *In particle physics every family of particles, a multiplet, has a shape: the points of its charges, its weights, span a polytope. For the symmetries SU(4) and SO(7) these shapes are exactly the solids of our cube–octahedron sweep, one multiplet per station with whole-number labels. The golden solids are never multiplets, because their numbers are not whole; they are where the Fibonacci multiplets converge, as the golden ratio is where the Fibonacci fractions converge.*

1. The theorem

Chapter 1 put the cube and the octahedron in one kaleidoscope (p = 4). Between them runs the sweep $S(\mu) = C \cap \mu C^*$ of #90 and #91: the cube $C = [-1, 1]^3$ cut by its polar octahedron grown by μ, from the octahedron (μ = 1) through the cuboctahedron (μ = 2) to the cube (μ = 3).

> **Theorem.** (a) The weight shape of the SU(4) multiplet with highest weight (a, b, a) — the multiplets that are their own antiparticle sets — is S(μ) with $\mu = (2a + b)/(a + b)$.
> (b) The weight shape of the SO(7) multiplet with highest weight (0, b, c) is S(μ) with $\mu = (4b + 3c)/(2b + c)$.
> (c) In four dimensions, the weight shape of the SO(9) multiplet (a, b, c, d) is the section $K \cap \mu K^*$ of the tesseract by its polar 16-cell (#91) with $\mu = \Sigma\lambda/\lambda_1$, $\lambda = (a+b+c+d/2, b+c+d/2, c+d/2, d/2)$.
> (d) The golden stations are limits of Fibonacci ladders: (F_{n+1}, F_n, F_{n+1}) of SU(4) → Icosaxes ($\mu = \varphi$); (F_n, F_{n+1}, F_n) → Unborn Axes ($\mu = 3 - \varphi$); $(0, F_n, F_{n+1}, 0)$ of SO(9) → the DICTO All-Golden Polytope with 8 Icosaxes cells ($\mu = \varphi^2$), and $(0, F_{n+1}, F_n, 0)$ → the one with 8 Unborn Axes cells ($\mu = 2 + 1/\varphi^2$). The Pell multiplets $(0, P_n, P_{n+1})$ of SO(7) converge to the Archimedean truncated cube ($\mu = 1 + \sqrt{2}$, the silver ratio).

Proof. A weight shape is the convex hull of the Weyl-group orbit of the highest weight, and it is the Minkowski sum of the fundamental shapes in proportion to the labels. For SU(4) the fundamental shapes are the tetrahedron T (ω_1), the octahedron O (ω_2) and −T (ω_3); with a = c the sum is $a(T + (-T)) + bO = a\cdot CO + b\cdot O$, whose walls are $x = \pm(a + b)$ (squares) and $\pm x \pm y \pm z = 2a + b$ (hexagons); scaling by $1/(a + b)$ gives $C \cap \mu C^*$. For SO(7) the orbit of $(b + c/2, b + c/2, c/2)$ under the signed permutations is the truncated cube with corners $(\pm1, \pm1, \pm s)$, $s = c/(2b + c)$, the section at $\mu = 2 + s$. For SO(9) the walls of the orbit of λ are $x_1 = \lambda_1$ and $|x|_1 = \Sigma\lambda$. (d): $\mu = \varphi$ needs $a/b = \varphi$; $\mu = 3 - \varphi$ needs $a/b = 1/\varphi$; $\mu = \varphi^2$ needs $c/b = \varphi$; $\mu = 1 + \sqrt{2}$ needs $c/b = \sqrt{2}$; the Fibonacci and Pell ratios are the best approximations of these. ∎

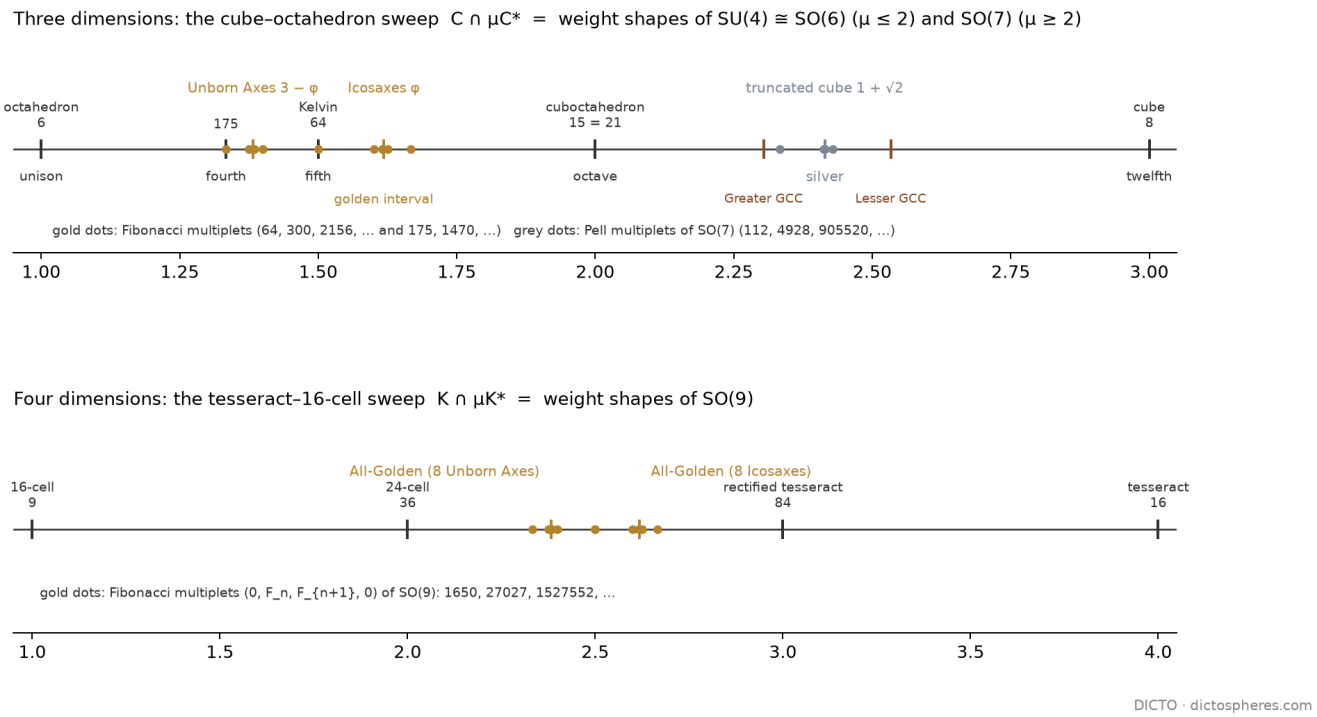


Figure 1. Above: the cube–octahedron sweep with its multiplets — the octahedron is the SU(4) sextet 6, Kelvin's truncated octahedron the 64, the cuboctahedron the 15 (and the SO(7) adjoint 21), the cube the SO(7) spinor 8; gold dots: Fibonacci multiplets closing in on Icosaxes and Unborn Axes; grey dots: Pell multiplets closing in on the truncated cube; the two Golden Cut Cubes (brown) have no ladder. Below: the four-dimensional sweep as SO(9) multiplets, with the Fibonacci ladders to the two DICTO All-Golden Polytopes.

2. The ladders

Ladder	Multiplets (dimension)	μ	Limit
SU(4) Fibonacci	64, 300, 2 156, 19 200, 218 700, 2 877 231	3/2, 5/3, 8/5, 13/8, 21/13, 34/21	Icosaxes, φ
SU(4) reversed	175, 1 470, 11 200, 127 575	4/3, 7/5, 11/8, 18/13	Unborn Axes, 3 − φ
SO(7) Pell	112, 4 928, 905 520, 411 098 688	7/3, 17/7, 41/17, 99/41	truncated cube, 1 + √2
SO(9) Fibonacci	1 650, 27 027, 1 527 552, 142 074 933	5/2, 8/3, 13/5, 21/8	All-Golden (Icosaxes cells), φ²
SO(9) reversed	15 444, 1 005 312, 77 144 760	7/3, 12/5, 19/8	All-Golden (Unborn Axes cells), 2 + 1/φ²

Along the SU(4) Fibonacci ladder the cubic tone of #94 fades to silence, 0.153, 0.050, 0.020, 0.008, 0.003, 0.001: the multiplets become more and more icosahedral, and only their limit, Icosaxes, is exactly so. The Golden Cut Cubes need label ratios 0.874… and 2.288… in the field of √2 and √5, with irregular continued fractions: no simple ladder reaches them, as #90's field obstruction leads one to expect.

3. One octave

For SU(4), μ = 1 + a/(a + b) takes every ratio between 1 and 2: the first half of the sweep is one octave, the octahedron its unison and the cuboctahedron its octave, and every just interval is a multiplet. The cube half (SO(7)) runs from the octave to the twelfth.

Interval	Ratio	Cents	Multiplet	Cubic tone Q_4
unison	1	0	6 (octahedron)	0.764
major third	5/4	386	384	0.552
perfect fourth	4/3	498	175	0.420
perfect fifth	3/2	702	64 (Kelvin)	0.153
minor sixth	8/5	814	2 156	0.020
golden interval	φ	833	limit: Icosaxes	**0**
major sixth	5/3	884	300	0.050
octave	2	1200	15 (cuboctahedron)	0.191

The cubic tone falls silent exactly at the golden interval, between the minor and the major sixth, whose multiplets are its first Fibonacci approximants. The least “whole” of all intervals is where the cubic order is fully hidden.

4. Known and new

Known: weight shapes as Weyl-group orbit hulls and their Minkowski additivity; the truncated octahedron as the permutohedron, the weight shape of SU(4)'s 64; the cuboctahedron and octahedron as the shapes of the 15 and the 6 (Georgi, *Lie Algebras in Particle Physics*; Humphreys, *Reflection Groups and Coxeter Groups*). **New here:** that these shapes are exactly the stations of the sweep with the stated μ, so that DICTO's named solids — Icosaxes, Unborn Axes and the DICTO All-Golden Polytopes — are the limits of Fibonacci ladders of multiplets, the truncated cube of a Pell ladder, and the Golden Cut Cubes of none; and the reading of the first half as an octave of just intervals, with the cubic tone silent at the golden interval.